

Progress Report

- Increment X -

Group Closet Design Game

Please use this template to describe your progress on the group project in the latest increment. Please do not change the font, font size, margins or line spacing. All the text in italic should be removed from your final submission.

1) Team Members

Ethan McConnell, edm22, ethanm177

Wesley Jordan, wsj21, Wefloo

Chloe Patrick, cjp22n, Chloe-Patrick

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2) Project Title and Description

Closet Design Game is a web-based single-page fashion design application that allows users to browse clothing templates by historical era and geographic region, place garments on a mannequin, and customize those garments with interactive drawing tools. The project is being built with React, Vite, Fabric.js, Tailwind CSS, and Lucide React. The longer-term scope includes multiple mannequin/body options, outfit layering, save/load functionality in browser storage, an outfit gallery, and export/sharing of completed designs as PNG images.

3) Accomplishments and overall project status during this increment

During Increment 1, the team established the core application structure and completed substantial portions of the planned studio interface, canvas tools, catalog architecture, and asset-processing pipeline.

- Canvas and drawing foundation: A React/Vite/Fabric.js canvas scaffold is in place. The current canvas supports freehand drawing, adjustable brush width, color selection, opacity control, and an eraser mode. A clipping boundary is applied to new freehand paths so strokes remain inside the current garment boundary. An FPS counter was also added to observe drawing performance.
- Studio user interface: The main branch contains a top navigation bar, era and region controls, clothing category controls, and a garment list driven by the clothing catalog. Users can filter available garments by era, region, and category, and the selected garment is stored in React state for canvas integration.
- Studio UI development: Completed the planned interface work for Increment 1, including era and region browsing controls, garment-category controls, responsive Studio UI elements, and the React state needed to support interaction with the canvas. The pair also contributed requirements and design documentation for the Studio UI portion of the project.
- Catalog and data architecture: A starter clothing catalog schema and JSON catalog are present. The catalog currently includes example garments with era, region, category, mannequin-frame target, SVG path, z-index, and anchor-coordinate fields.
- SVG asset pipeline: A Python normalization script was created to scale and position raw SVG clothing assets into a common 500x800 coordinate system using category-specific anchor points. A normalized 1990s Harajuku skirt asset is included as the first processed garment.

- Pipeline bug resolution: The team identified and fixed a duplicate XML namespace issue that caused normalized SVG files to fail to render correctly in browsers and editors.

Overall, the team completed the assigned Increment 1 coding work across the canvas, Studio UI, and data/asset pipeline. Together, these pieces establish the project's core foundation and provide the components needed for the more advanced customization, saving, layering, and gallery features planned for Increment 2.

4) Challenges, changes in the plan and scope of the project and things that went wrong during this increment

- Integration and version-control coordination: Because multiple project areas were developed in parallel, the team had to coordinate branch changes and resolve a small number of merge/integration conflicts before combining work.
- Asset consistency: Normalizing garment SVGs so they rendered and aligned consistently required some adjustment to the asset-processing pipeline.
- Component coordination: The canvas, Studio UI, and data pipeline were developed as separate components, so the team needed to make sure their interfaces and shared data structures stayed compatible as the increment progressed.
- Testing and refinement: As features were combined, the team spent additional time checking that the UI, canvas behavior, catalog data, and normalized assets worked together as expected.

These challenges did not require a major change to the project scope. The team remained on track with the planned Closet Design Game functionality and addressed issues as they appeared during development.

5) Team Member Contribution for this increment

Chloe Patrick

- Progress report: Wrote and organized Progress Report 1, including the project-status summary and stakeholder communication.
- Requirements and design: Contributed to the Requirements and Design (RD) document with Camryn, including requirements/design content for the Studio UI portion of the project.
- Implementation/testing document: Contributed to the Implementation and Testing (IT) document for Increment 1.
- Source code: As part of Pair 2 with Camryn, completed the assigned Studio UI code, including the era/region interface, clothing-category controls, React selection state, responsive interface elements, and the UI handoff needed to work with the canvas.

Camryn Morris

- Requirements and design: Contributed to the Requirements and Design (RD) document for Increment 1, working on the Studio UI requirements/design material with Chloe.
- Source code: As part of Pair 2 with Chloe, completed the assigned Studio UI code. Camryn also contributed the Studio UI navigation/filter implementation that was merged into the main branch, including era/region selectors, category controls, catalog-driven garment filtering, and garment-selection state.

Ethan McConnell

- Source code: As part of Pair 1 with Jessika, completed the assigned canvas and drawing-tool code for Increment 1, including the Fabric.js canvas foundation and brush-control functionality.
- Video/presentation: Contributed to the Increment 1 video/presentation.

Jessika Tyler

- Source code: As part of Pair 1 with Ethan, completed the assigned canvas code for Increment 1, including boundary-clipping behavior that keeps drawing strokes within the active garment area and performance/FPS tracking.
- Video/presentation: Contributed to the Increment 1 video/presentation.

Vanessa Campoe

- Source code: As part of Pair 3 with Wesley, completed the assigned data and asset-pipeline code for Increment 1, including the clothing-catalog schema/starter data, SVG normalization pipeline, garment assets, and pipeline bug fixes.
- Video/presentation: Contributed to the Increment 1 video/presentation.

Wesley Jordan

- Implementation/testing document: Contributed to the Implementation and Testing (IT) document for Increment 1.
- Source code: As part of Pair 3 with Vanessa, completed the assigned data and pipeline code for Increment 1 and contributed to repository integration/review work around the catalog architecture.
- Video/presentation: Completed part of the Increment 1 video/presentation.

6) Plans for the next increment

Increment 2 will focus on expanding the current foundation into a more complete outfit-design workflow. The team plans to complete the following Increment 2 work:

- Implement reliable layer ordering so the mannequin remains behind clothing, bottoms remain under tops when appropriate, and accessories stay on top.
- Add mannequin configuration controls for different frames/body options and connect those controls to the canvas without losing the user's current outfit.
- Extend clipping behavior so changing garments also updates the active drawing boundary without breaking existing painted layers.
- Implement client-side save/load/delete functionality using LocalStorage or IndexedDB.
- Create the Saved Outfits Gallery interface with thumbnail cards and Load/Delete actions.
- Add undo/redo history for brush strokes and clothing changes and stress-test the history stack.
- Serialize the Fabric.js canvas state to JSON and restore it accurately, including garments, layer order, and brush paths.
- Add integration tests for saving, loading, deleting, and layer ordering; increase automated test coverage and document results in the IT deliverable.
- Complete/polish the remaining Requirements and Design diagrams and descriptions and prepare the Increment 2 progress report and demonstration video.

7) Stakeholder Communication

Hello,

The Closet Design Game has reached a strong foundation for its design studio. The application now includes the core web interface, historical era and regional clothing filters, category browsing, and an interactive drawing canvas with adjustable brush width, color, opacity, and eraser controls. We have also established the clothing catalog structure and an automated process for resizing and aligning garment artwork to a consistent canvas format.

During this development cycle, the team also coordinated work across the interface, canvas, and asset pipeline and made the adjustments needed to keep those components compatible. These refinements strengthened the project foundation without changing the intended scope.

With the Increment 1 foundation in place, the next phase will build on the completed canvas, interface, and catalog work to support a fuller outfit-design experience.

Next, we will build the features that make the studio feel like a complete outfit-design experience: dependable garment layering, mannequin customization, undo/redo, local outfit saving and loading, and a saved-outfit gallery. We will continue testing these features as they are integrated so that later catalog expansion and export/sharing work can build on a stable base.

Best,

Closet Design Game Team

8) Link to video

<https://drive.google.com/file/d/1jvS5ugH7LDweYcBRr1QIzFTwebNgjDNQ/view?usp=sharing>